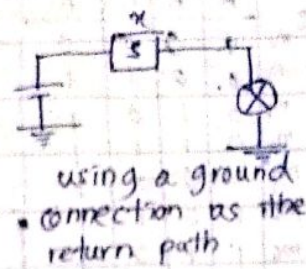
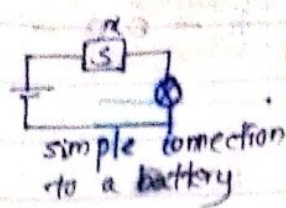
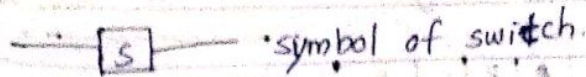
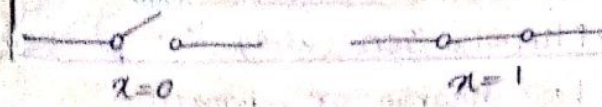


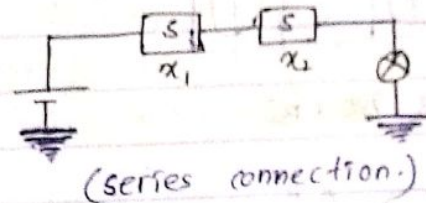
Digital Logic Design



*) $L(x_1, x_2) = x_1 \cdot x_2$

where $L=1$ if $x_1=1$ and $x_2=1$
otherwise $L=0$

* This ' $\cdot$ ' symbol is called AND operator.

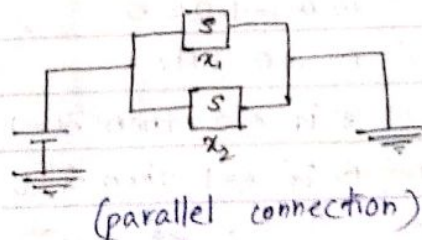


*) $L(x_1, x_2) = x_1 + x_2$

where $L=1$ if $x_1=1$ or $x_2=1$ or
if $x_1=x_2=1$

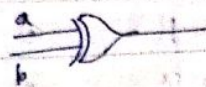
$L=0$ if $x_1=x_2=0$

* This '+' symbol is called OR operator.



XOR gate.

a	b	a XOR b
0	0	0
0	1	1
1	0	1
1	1	0



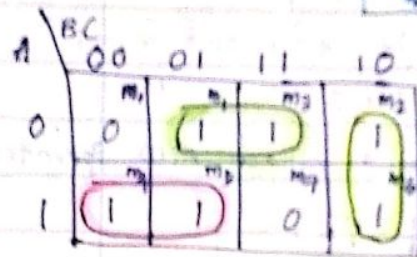
$a \oplus b$

* Designing combinational circuit

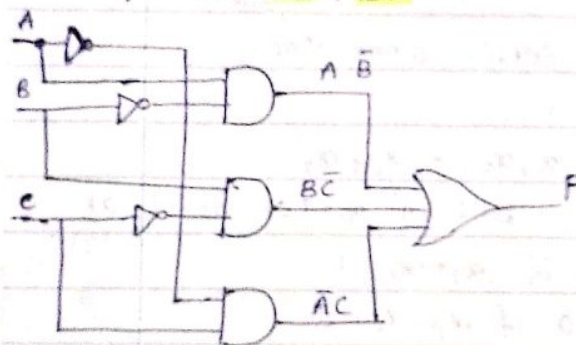
- (1) Truth table
- (2) Minimization (using K-map)
- (3) Logic diagram or schematic

Exs

	A	B	C	F
m_0	0	0	0	0
m_1	0	0	1	1
m_2	0	1	0	1
m_3	0	1	1	1
m_4	1	0	0	1
m_5	1	0	1	1
m_6	1	1	0	1
m_7	1	1	1	0



$$F = \overline{A}B + \overline{A}C + BC$$



Axioms of Boolean Algebra.

- ⊙ $0 \cdot 0 = 0$
- ⊙ $0 \cdot 1 = 1 \cdot 0 = 0$
- ⊙ $1 + 1 = 1$
- ⊙ $1 + 0 = 0 + 1 = 1$
- ⊙ $1 \cdot 1 = 1$
- ⊙ if $x = 0$, then $\overline{x} = 1$
- ⊙ $0 + 0 = 0$
- ⊙ if $x = 1$, then $\overline{x} = 0$

Single variable Theorems.

- ⊙ $x \cdot 0 = 0$
- ⊙ $x + \overline{x} = 1$
- ⊙ $x \cdot 1 = x$
- ⊙ $x + 0 = x$
- ⊙ $x \cdot x = x$
- ⊙ $x + x = x$
- ⊙ $x \cdot \overline{x} = 0$
- ⊙ $x + \overline{x} = 1$
- ⊙ $\overline{\overline{x}} = x$

Two and Three variable properties:

- ① $x \cdot y = y \cdot x \rightarrow$ commutative.
- ② $x + y = y + x \rightarrow$ commutative
- ③ $x \cdot (y \cdot z) = (x \cdot y) \cdot z$
- ④ $x + (y + z) = (x + y) + z$ } $\rightarrow$ Associative
- ⑤ $x \cdot (y + z) = x \cdot y + x \cdot z$
- ⑥ $x + y \cdot z = (x + y) \cdot (x + z)$ } $\rightarrow$ Distributive
- ⑦ $x + x \cdot y = x$
- ⑧ $x \cdot (x + y) = x$ } $\rightarrow$ Absorption.
- ⑨ $x \cdot y + x \cdot \bar{y} = x$
- ⑩ $(x + y) \cdot (x + \bar{y}) = x$ } combining.
- ⑪ $\bar{x} \cdot y = \bar{x} + \bar{y}$
- ⑫ $\bar{x} + \bar{y} = \overline{x \cdot y}$ } De Morgan's theorem
- ⑬ $x + \bar{x} \cdot y = x + y$
- ⑭ $x \cdot (\bar{x} + y) = x \cdot y$
- ⑮ $x \cdot y + y \cdot z + \bar{x} \cdot z = x \cdot y + \bar{x} \cdot z$
- ⑯ $(x + y) \cdot (y + z) \cdot (\bar{x} + z) = (x + y) \cdot (\bar{x} + z)$ } consensus

Sum of products (SOP)

Row no.	x_1	x_2	x_3	Minterm	Maxterm
0	0	0	0	$m_0 = \bar{x}_1 \bar{x}_2 \bar{x}_3$	$M_0 = x_1 + x_2 + x_3$
1	0	0	1	$m_1 = \bar{x}_1 \bar{x}_2 x_3$	$M_1 = x_1 + x_2 + \bar{x}_3$
2	0	1	0	$m_2 = \bar{x}_1 x_2 \bar{x}_3$	$M_2 = x_1 + \bar{x}_2 + x_3$
3	0	1	1	$m_3 = \bar{x}_1 x_2 x_3$	$M_3 = x_1 + \bar{x}_2 + \bar{x}_3$
4	1	0	0	$m_4 = x_1 \bar{x}_2 \bar{x}_3$	$M_4 = \bar{x}_1 + x_2 + x_3$
5	1	0	1	$m_5 = x_1 \bar{x}_2 x_3$	$M_5 = \bar{x}_1 + x_2 + \bar{x}_3$
6	1	1	0	$m_6 = x_1 x_2 \bar{x}_3$	$M_6 = \bar{x}_1 + \bar{x}_2 + x_3$
7	1	1	1	$m_7 = x_1 x_2 x_3$	$M_7 = \bar{x}_1 + \bar{x}_2 + \bar{x}_3$

SOP =

Row no	f	Row no	f
0	0	4	1
1	1	5	1
2	0	6	1
3	0	7	0

* Here,

$$SOP = m_1 + m_4 + m_5 + m_6$$

(2) $f(x_1, x_2, x_3) = \bar{x}_1 \bar{x}_2 x_3 + x_1 \bar{x}_2 \bar{x}_3 + x_1 \bar{x}_2 x_3$
 $+ x_1 x_2 \bar{x}_3$

Ex:

$$= (\bar{x}_1 + x_1) \bar{x}_2 x_3 + x_1 (\bar{x}_2 + x_2) \bar{x}_3$$

$$= 1 \cdot \bar{x}_2 x_3 + x_1 \cdot 1 \cdot \bar{x}_3$$

$$= \bar{x}_2 x_3 + x_1 \bar{x}_3$$

$$SOP = \sum (m_1, m_4, m_5, m_6)$$

$$= \sum m(1, 4, 5, 6)$$

Product of Sum (POS)

$$POS = f = M_0 \cdot M_2 \cdot M_3 \cdot M_7$$

$$= (x_1 + x_2 + x_3) \cdot (x_1 + \bar{x}_2 + x_3) \cdot (x_1 + \bar{x}_2 + \bar{x}_3) \cdot (x_1 + x_2 + \bar{x}_3)$$

$$= (x_1 + x_3) (\bar{x}_2 + \bar{x}_3)$$

A
B

$$POS = \prod (M_0, M_2, M_3, M_7)$$

$$= \prod M(0, 2, 3, 7)$$

* The "TT" sign denotes the logical product operation.

		$x_1 x_2$		$\bar{x}_3 \bar{x}_4$		
		00	01	11	10	
$x_3 x_4$	00	1	1	1	1	$x_1 \bar{x}_2 \bar{x}_4$
	01			1		$x_1 x_2 \bar{x}_3$
	11			1	1	$x_1 x_2 x_4$
	10				1	$x_1 x_3 x_4$
						$x_1 \bar{x}_2 x_3$

radix = base of the number system.

conjunction = AND = $\wedge$ = $\cdot$

disjunction = OR = $\vee$ = $+$

Negation = NOT = $\neg$ = $'$!

Example:

Q. Consider the an arbitrary number system with the independent digits as 0, 1 and X. what is the radix of this number system? list the first 10 numbers in this number system.

- radix is 3.

- 0, 1, X, 10, 11, 1X, X0, X1, XX, 100

(first 16 numbers in binary number system.)

0, 1, 10, 11, 100, 101, 110, 111, 1000, 1001, 1010, 1100, 1101, 1110, 1111

- 1's complement of 10010110_2 is 01101001_2

- 2's complement of a binary number is obtained by adding '1' to its 1's complement.

2's complement of 10010110_2 is,

01101001

$\begin{array}{r} 1 \\ \hline (01101001)_2 \end{array}$

- In decimal number system we have 9's and 10's complement.

the 9's complement of a given decimal number is obtained by subtracting each digit from 9.

9's complement of $(2496)_{10}$ is

$(7503)_{10}$

10's complement $\rightarrow$ add '1' to 9's comp^{*}

7503

$\begin{array}{r} 1 \\ \hline (7504)_{10} // \end{array}$

* In octal number system we have 7's and 8's complements.

• 7's complement of a given octal number is obtained by subtracting each octal digit from 7.

$$E) \quad 562_8 \rightarrow 7's \text{ complement} \rightarrow 215_8$$

$$562_8 \rightarrow 8's \text{ complement} \rightarrow 216_8$$

15's complement of Hexadecimal number can be obtained by subtracting each digit from 15.

$$(3BF)_{16} \rightarrow 15's \text{ complement} \rightarrow (C40)_{16}$$

$$(3BF)_{16} \rightarrow 16's \text{ complement} \rightarrow (C41)_{16}$$

Sign-Bit Magnitude

'0' denoting '+'

'1' denoting '-'

eight bit representation of +9,

00001001

eight bit representation of -9,

10001001

Binary addition & subtraction.

$$0+0=0$$

$$0+1=1$$

$$1+0=1$$

$$1+1=0 \text{ with a carry of '1' to}$$

the next more significant bit

$$1+1+1=1 \text{ with a carry of '1' to}$$

the next more significant bit

$$0-0=0$$

$$0-1=1 \text{ with a borrow of 1 from}$$

next more significant bit

$$1-0=1$$

$$1-1=0$$

Binary Addition of three bits

A	B	C _{in}	Sum	Count
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Binary subtraction of three bits

Minuend A	Subtrahend B	Borrow-in B _{in}	D (Difference)	Bout.
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Addition using 2's complement method

1. Both the numbers are positive.

+37 and +18

$$\begin{array}{r}
 00100101_2 \\
 \downarrow 2's \\
 00100101
 \end{array}
 \quad
 \begin{array}{r}
 00010010_2 \\
 \downarrow 2's \\
 00010010
 \end{array}$$

$$\begin{array}{r}
 00100101 \\
 + 00010010 \\
 \hline
 00110111
 \end{array}$$

$$\begin{array}{r}
 0 + 0 + 32 + 16 + 0 + 4 + 2 + 1 \\
 + 55 \\
 \hline
 \end{array}$$

ANS

2. Larger of the two numbers is positive

+37 and +18

00100101 00010010

↓ 2's

↓ 2's

00100101

11101101

+1

11101110 + 2's

00100101

+ 11101110

1 00010011

Carry

16 + 2 + 1 = 19

3. The larger of the two numbers is negative.

+18 and -37

00010010

00100101

↓ 2's

↓ 2's

00010010

11011010

11011011

00010010

+ 11011011

11101101

~~representant in sign~~

-19

Both the numbers are negative.

-18 Date: and -37
 00010010
 $\downarrow$ 2's

~~11101101~~

11101101

$\downarrow$ 2's

11101101

11101110

00100101
 $\downarrow$ 2's

11011010

$\downarrow$ 2's

11011010

11011011

11011011

11101110

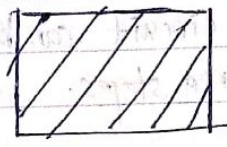
+ 11011011

111001001

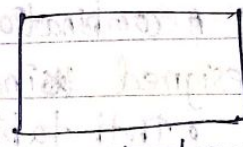
= -55

carry

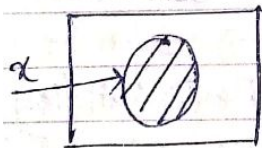
Venn diagram representations.



constant 1



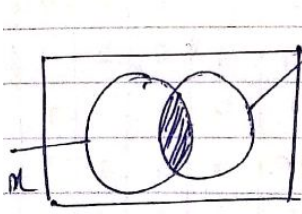
constant 0



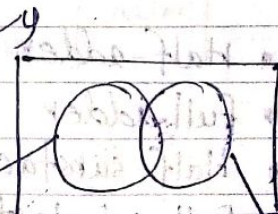
x



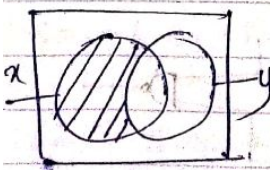
$\bar{x}$



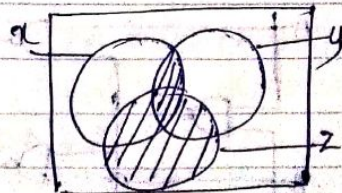
$x \cdot y$



$x + y$

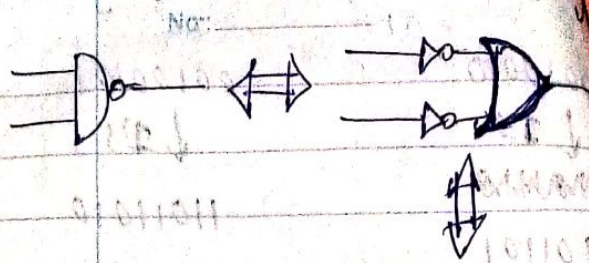


$x \cdot \bar{y}$

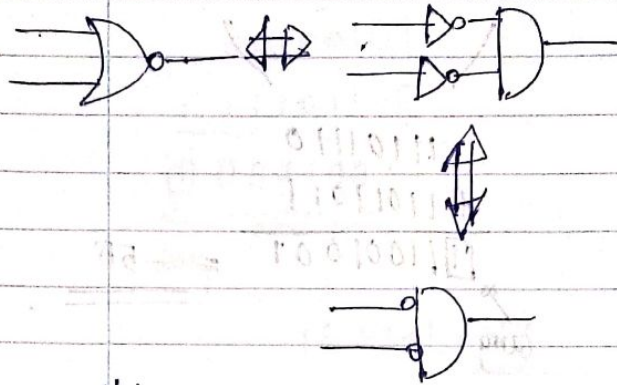


$x \cdot y + z$

De Morgan's Theorem in Logic



(a) $\overline{xy} = \overline{x} + \overline{y}$



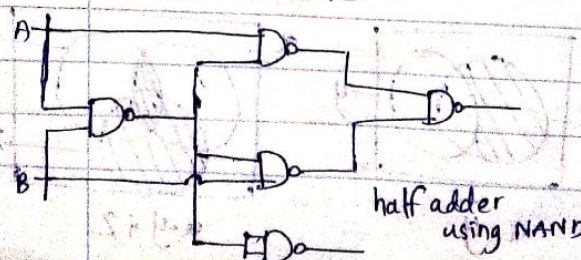
(b) $\overline{x+y} = \overline{x} \cdot \overline{y}$

* A combinational circuit can be designed using three steps.

- Truth table
- Minimization.
- Logic diagram or schematic.

Arithmetic circuits-Basic Building Blocks

- Half-adder
- Full-adder
- Half-subtractor
- Full-subtractor
- Controlled inverter.



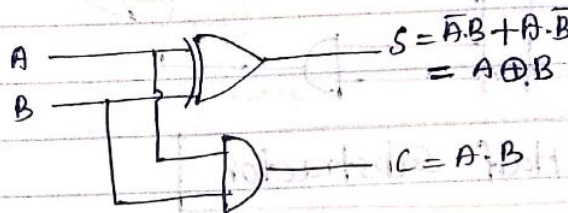
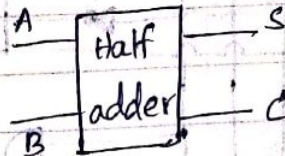
Half-adder

This is an arithmetic circuit block that can be used to add two bits.

$$\text{SUM} = A \cdot \bar{B} + \bar{A} \cdot B$$

$$\text{Carry } C = A \cdot B$$

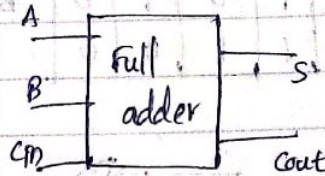
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



Full-adder.

This circuit is an arithmetic circuit block that can be used to add three bits to produce a sum and a CARRY output.

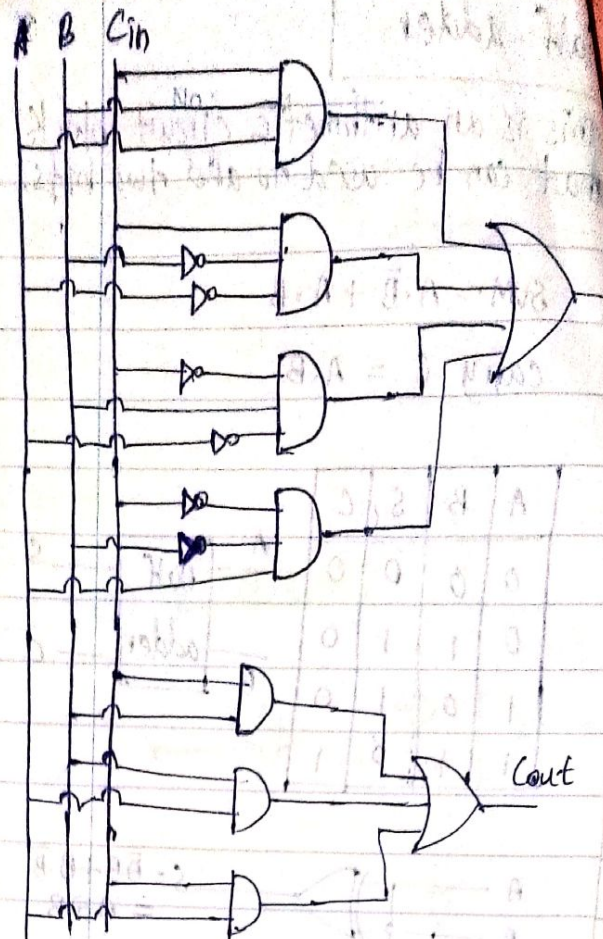
A	B	C _{in}	S	C _{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



$$S = \bar{A} \cdot \bar{B} \cdot C_{in} + \bar{A} \cdot B \cdot \bar{C}_{in} + A \cdot \bar{B} \cdot \bar{C}_{in} + A \cdot B \cdot C_{in}$$

$$C_{out} = B \cdot C_{in} + A \cdot B + A \cdot C_{in}$$

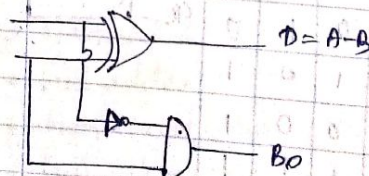
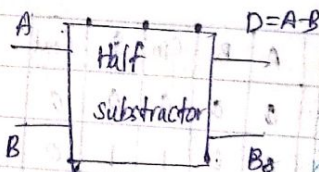
Atlas



Half Subtractor

* This is a combinational circuit that can be used to subtract one binary digit from another to produce a DIFFERENCE output and a BORROW output.

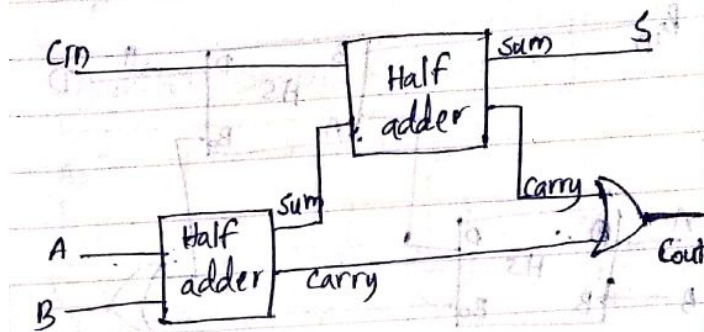
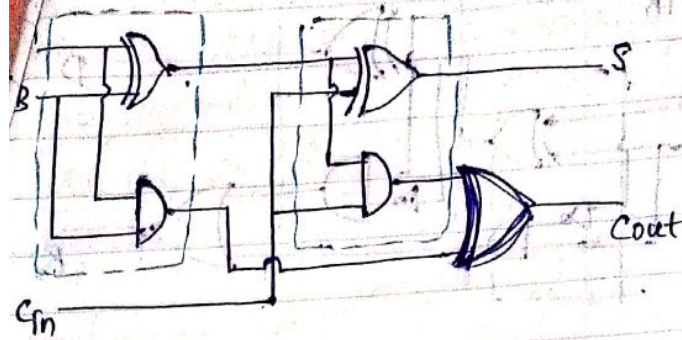
A	B	D	B ₀
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0



$$D = \bar{A} \cdot B + A \cdot \bar{B}$$

$$B_0 = \bar{A} \cdot B$$

logical implementation of a full adder with half-adders.



Full Subtractor

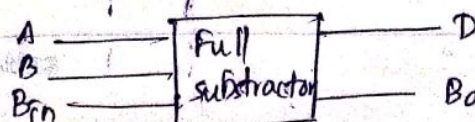
* A full subtractor performs subtraction whether a '1' has already been borrowed by the previous adjacent lower minuend bit or not.

$$D = \bar{A}\bar{B}B_{in} + \bar{A}BB_{in} + A\bar{B}\bar{B}_{in} + AB\bar{B}_{in}$$

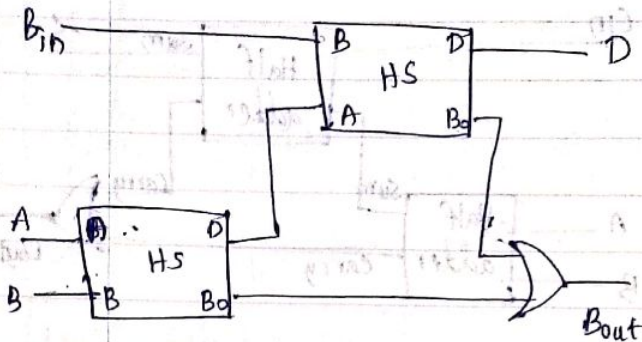
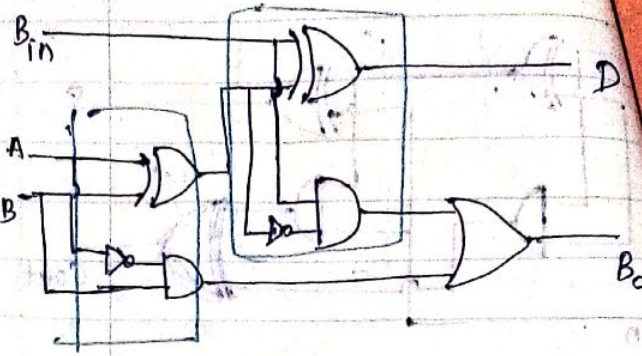
$$B_o = \bar{A}\bar{B}B_{in} + \bar{A}BB_{in} + A\bar{B}B_{in} + AB\bar{B}_{in}$$

$$B_o = \bar{A}B + A\bar{B}_{in} + B \cdot B_{in}$$

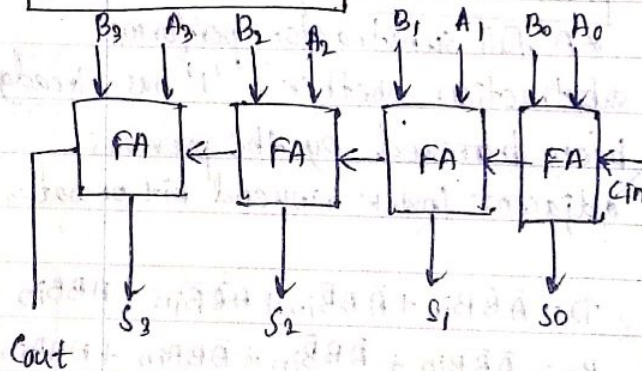
A	B	B_{in}	D	B_o
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1



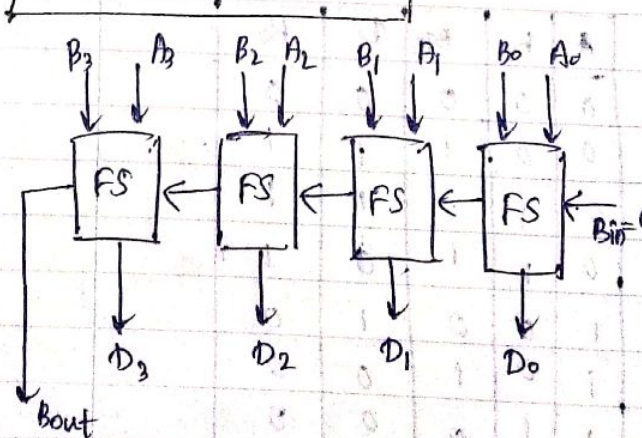
Logic implementation of a full subtractor with half-subtractor



Four bit adder



Four bit subtractor

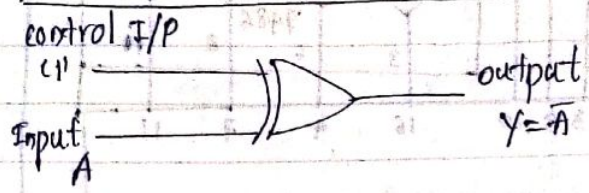


Controlled Inverter

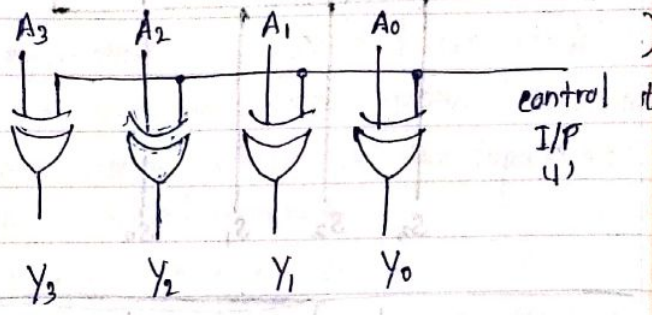
used to find 1's complement.

* A controlled inverter is needed when an adder is to be used as a subtractor.

one bit controlled inverter

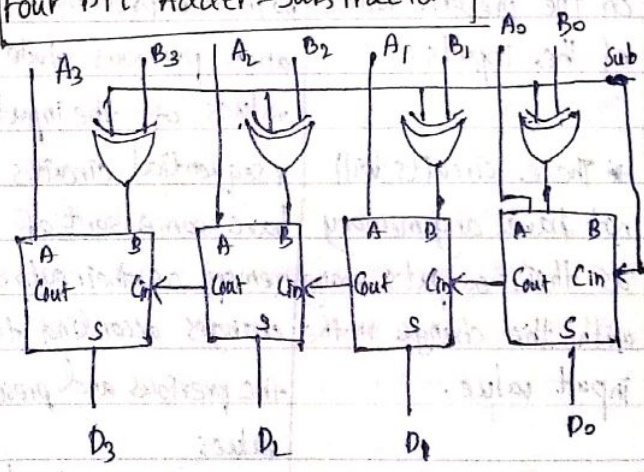


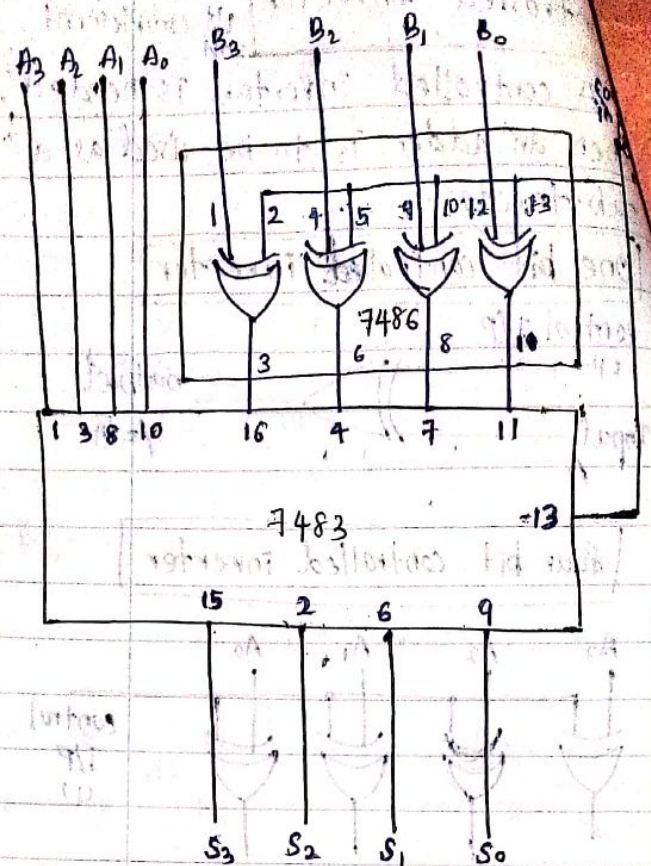
four bit controlled inverter.



11010010 → 00101101

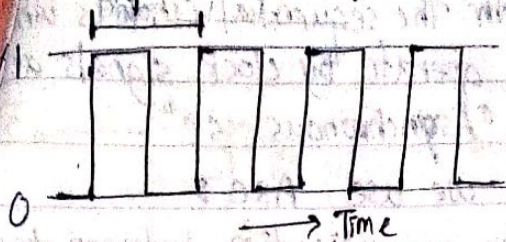
Four bit Adder-Subtractor





combinational circuit	sequential circuits
* Output depends only on the present value of the inputs. * These circuits will not have any memory as their outputs change with the change in the input value. * There are no feedback involved. * They are used in basic Boolean operations. * Implemented in Half adder, full adders, multiplexers, demultiplexers, decoders and encoders.	* Output depends on both the present and previous state values of the inputs. * sequential circuits have some sort of memory as their output changes according to the previous and present values. * In a sequential circuit the outputs are connected to it as a feedback path. * used in the designing of memory devices. * Implemented in, RAM, registers, counters and other state retaining machines.

Clock signal in sequential circuits



Sequential circuits

↓
synchronous sequential circuits Asynchronous sequential circuits.

• Synchronous sequential circuits (SSC)

In SSC, the output depends on present and previous states of the inputs at the clocked instances.

• Where we use SSC?

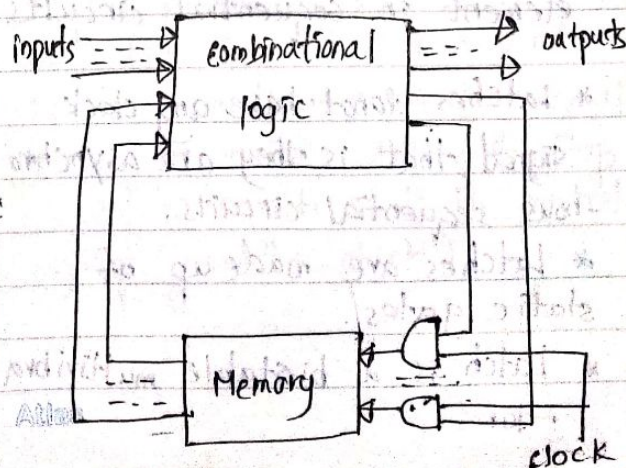
* In the design of MOORE-MEALY state management machines.

* They are used in synchronous counters, flipflops, etc.

Limitations:

• All the flipflops in SSC must be connected to clock signal.

* • Critical path or the slowest path determines the maximum possible clock frequency. Hence they are slower than asynchronous circuits.



• Asynchronous sequential circuits
* The sequential circuits which do not operate by clock signals are called "Asynchronous SC".

where we use 'ASC'.

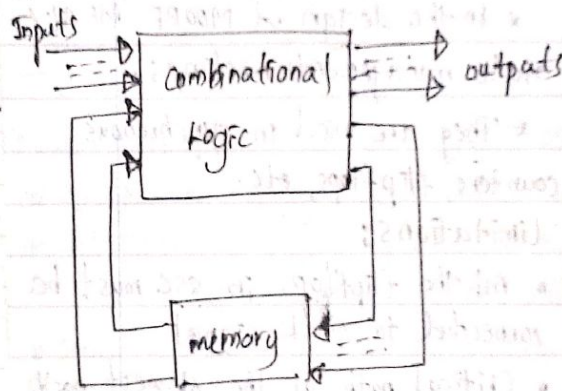
* in the communication between two units having their own independent clocks.

* when we require the better external input handling.

Limitations.

• Asynchronous sequential circuits are more difficult to design.

* Though they have a faster performance, their output is uncertain.



Latch

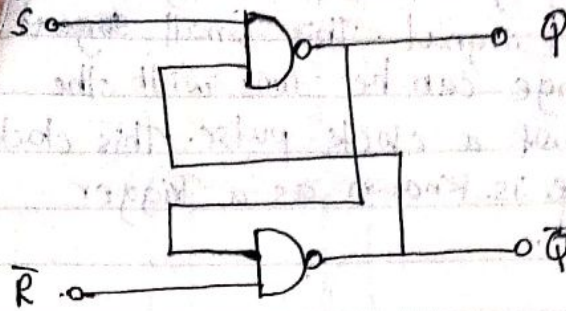
* Latch is a basic building element in sequential circuits.

* Latches do not have any clock signal, that is they are asynchronous sequential circuits.

* Latches are made up of static gates.

* Latch is a bistable multivibrator.

S-R Latch

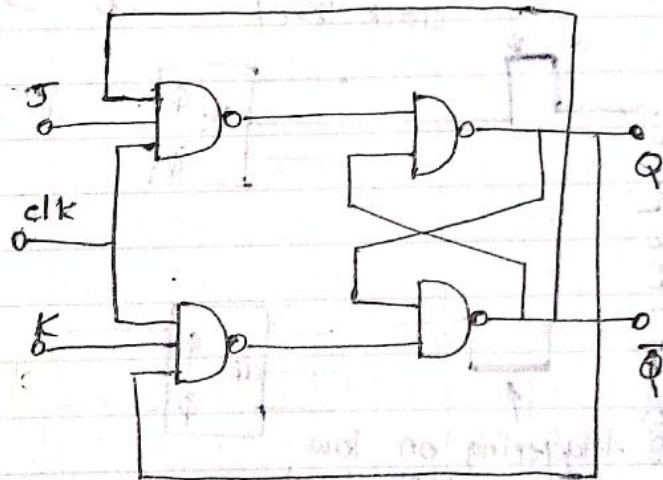


FlipFlop

* This also a building block of ~~seq~~ synchronous sequential circuits.

- has two stable states
- can store one bit of information

J-K flipflop.



* only difference between latches and flip-flops is that a latch is level sensitive to control signal (clock or enable) while a flipflop is edge sensitive to the control signal (usually clock)

Triggering: The change in out of a flip-flop can be done by bringing a small change in the input signal. This small ~~signal~~ change can be done with the help of a clock pulse. This clock pulse is known as a Trigger pulse.

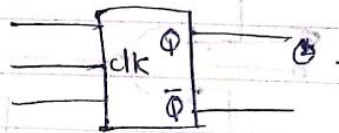
* flip-flops are basic components in registers and counters.

Triggering
 ↳ level triggering
 ↳ edge triggering

level triggering
 ↳ High level triggering
 ↳ low level triggering

triggering on high clock level

Triggers on this edge of clock pulse.



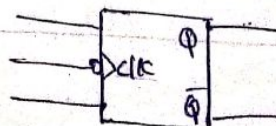
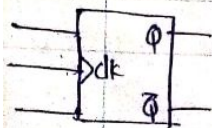
triggering on low clock level.



Edge triggering

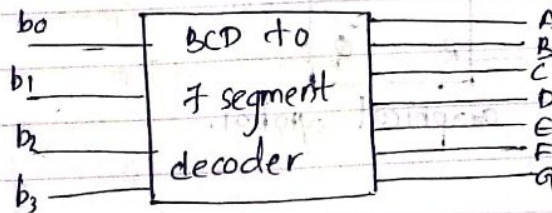
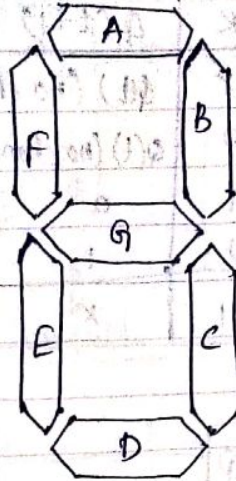
positive edge triggering

negative edge triggering



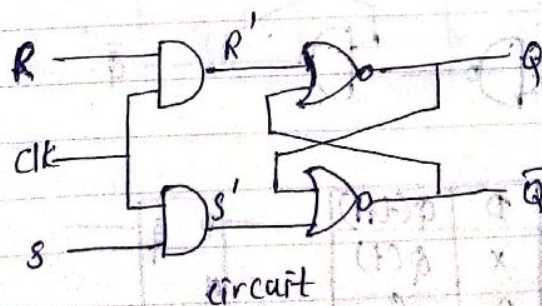
Seven segment Decoder.

Date: / /



	b_0	b_1	b_2	b_3	A	B	C	D	E	F	G
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	1	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1

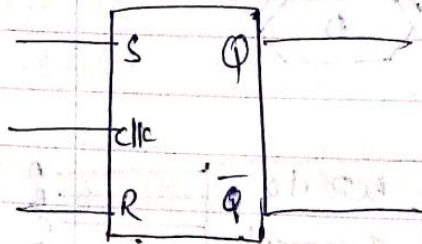
Mastered SR Latch.



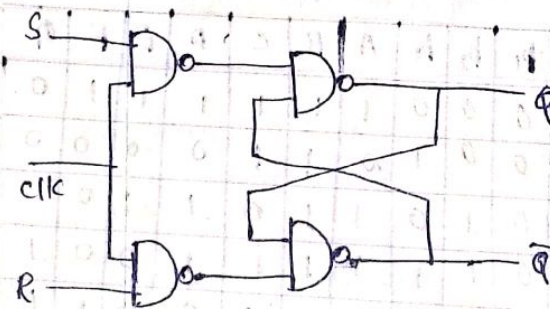
Atlas

characteristic table for SR

clk	S	R	$Q(t+1)$
0	x	x	$Q(t)$ (no change)
1	0	0	$Q(t)$ (no change)
1	0	1	0
1	1	0	1
1	1	1	x

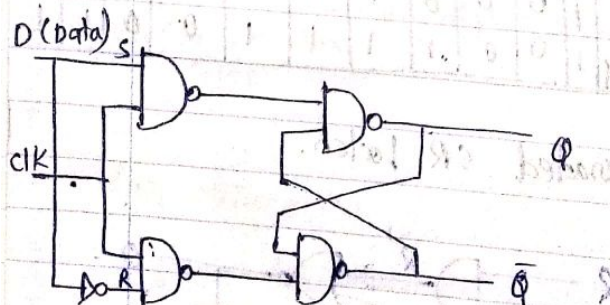


graphical symbol.



Gated SR latch with NAND gates.

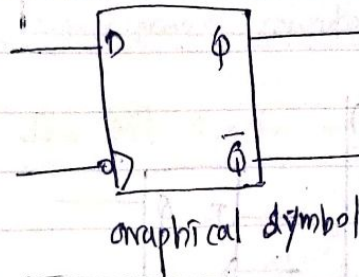
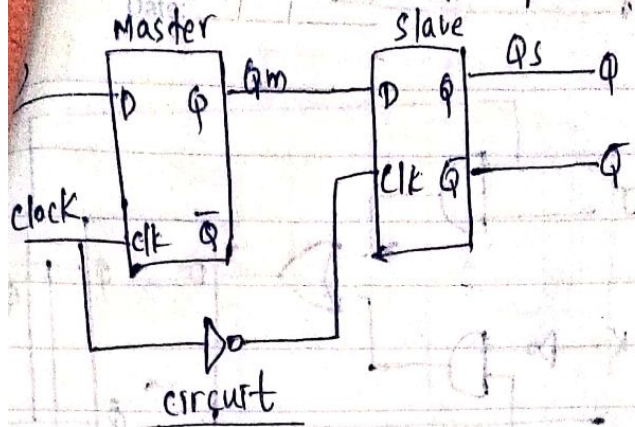
Gated D Latch



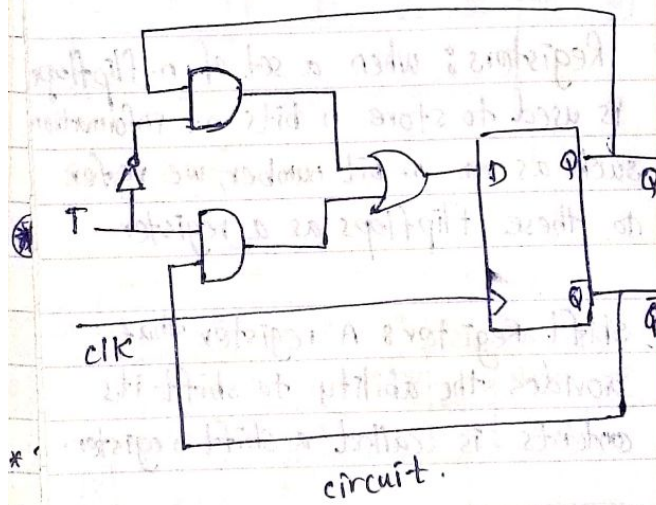
clk	D	$Q(t+1)$
0	x	$Q(t)$
1	0	0
1	1	1



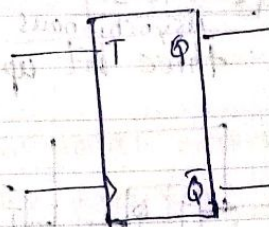
edge triggered D flip flop



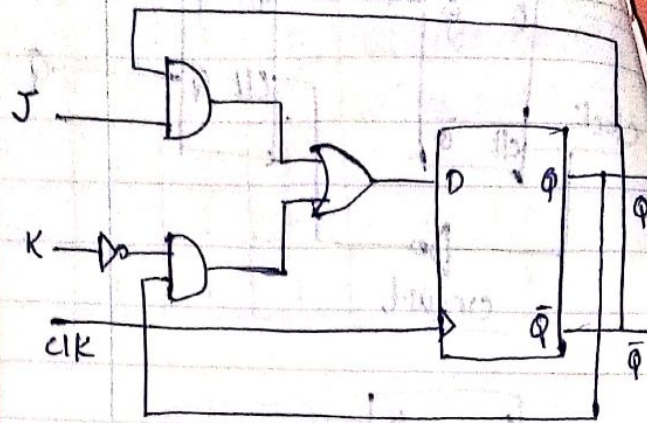
T-Flipflop



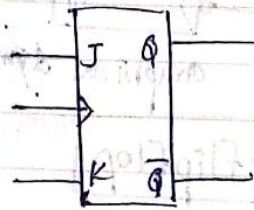
T	$Q(t+1)$
0	$Q(t)$
1	$\bar{Q}(t)$



J-K flipflops



J	K	$Q(t+1)$
0	0	$Q(t)$
0	1	0
1	0	1
1	1	$\bar{Q}(t)$

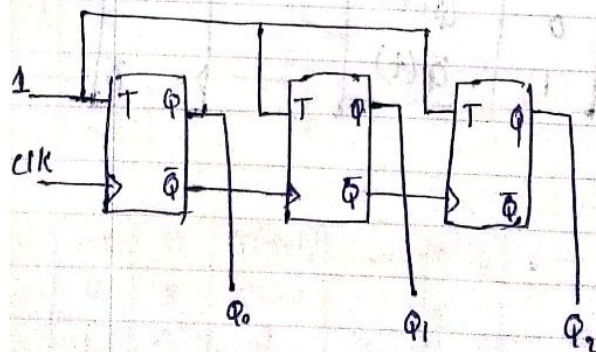


Registers: When a set of n -flipflops is used to store n bits of information, such as an n -bit number, we refer to these flipflops as a register.

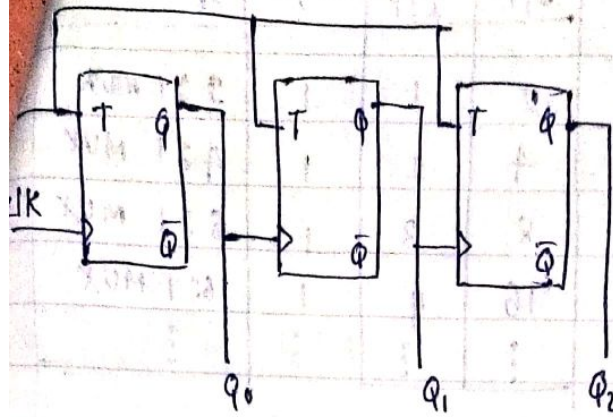
Shift Register: A register that provides the ability to shift its contents is called a shift register.

Counters

Asynchronous counters
three bit up counter.

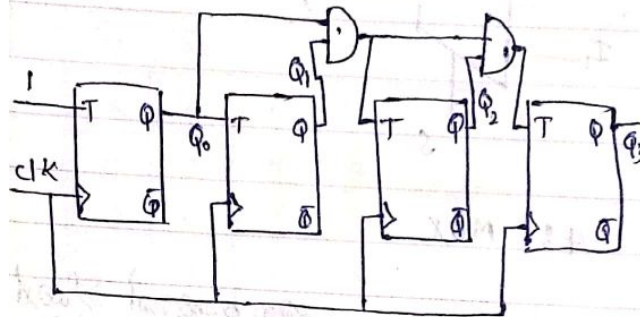


3 bit down counter..

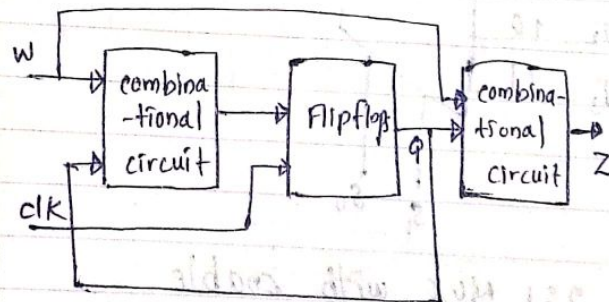


— synchronous counters —

four bit synchronous up counter.



General form of a sequential circuit



Multiplexers

* It is a combinational circuit that selects binary data from one of many inputs lines and directs it to the output line.

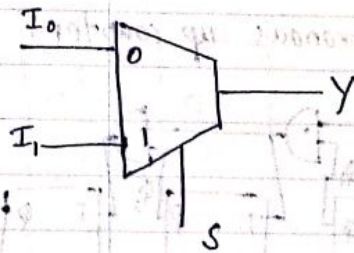
Inputs - 2^n

output - 1

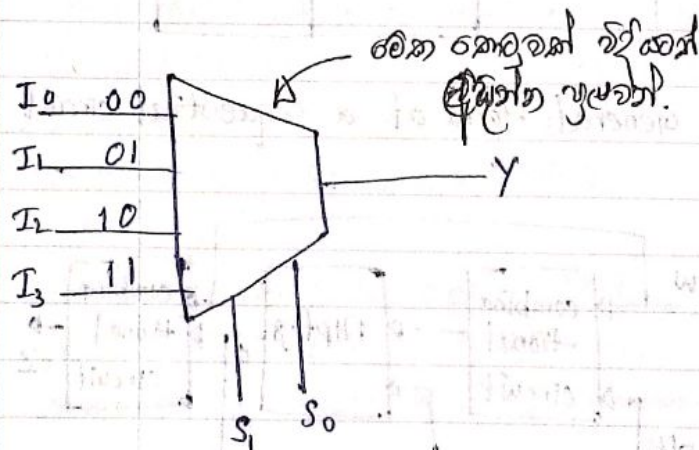
selection inputs - n

I/P	Selection I/P	O/P	Type of MUX
2	1	1	2:1 MUX
4	2	1	4:1 MUX
8	3	1	8:1 MUX
16	4	1	16:1 MUX
⋮	⋮	⋮	⋮

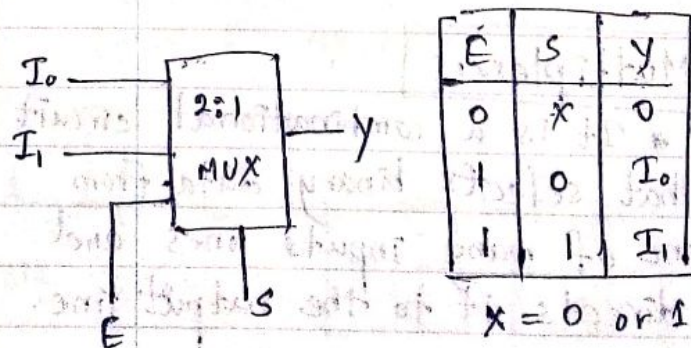
2:1 MUX (2:1 line multiplexer)



4:1 MUX

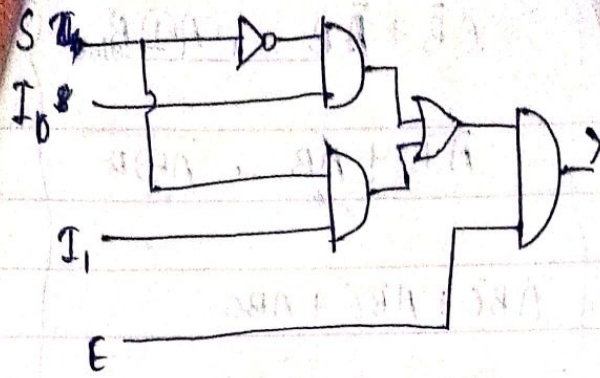


2:1 MUX with enable



$$Y = E \cdot \bar{S} \cdot I_0 + E \cdot S \cdot I_1$$

$$= E \cdot (\bar{S} \cdot I_0 + S \cdot I_1)$$



Example Question

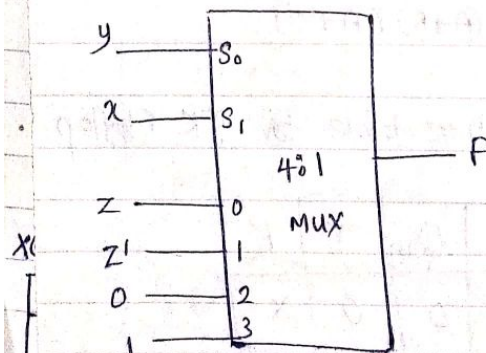
$$F(x, y, z) = \sum(1, 2, 6, 7)$$

the values for the data input $\rightarrow$ from the truth table of the function.

$S_1 \quad S_0$

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

$F = Z$ (for row 0,1)
 $F = Z'$ (for row 0,2)
 $F = 0$ (for row 0,3)
 $F = 1$ (for row 0,4)



$$A\bar{B} + \bar{A}B = A \oplus B$$

$$\bar{A}\bar{B} + AB = \overline{A \oplus B}$$

$$\bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + ABC$$

$$= \overline{\bar{A}\bar{B}\bar{C} \cdot A\bar{B}\bar{C} \cdot ABC}$$

$$(A+B+\bar{C})(\bar{A}+B+\bar{C})(\bar{A}+\bar{B}+C)$$

$$= \bar{A}\bar{B}\bar{C} \cdot A\bar{B}\bar{C} \cdot A\bar{B}\bar{C}$$

Redundancy theorem.

I) Three variables.

II) Each variable is repeated twice.

III) One variable is complimented.

IV) Take the complimented variable.

Examples

$$\begin{aligned} \textcircled{1} \quad & AB + A'C + BC \\ &= \underline{AB + A'C} \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad & (A+B)(\bar{A}+C)(B+C) \\ &= (A+B)(\bar{A}+C) \end{aligned}$$

excitation table for JK Flipflop

Q	Q _{next}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0